

# **DETECTION OF FAKE AND AI-GENERATED PRODUCT REVIEWS USING DEEP LEARNING**

**Synopsis report on  
Deep Learning Project  
[ICT-4442]**

Submitted By

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## PART A: SYNOPSIS (PHASE 1)

### Team Details:

| S No. | Name                   | Reg. No.  | Branch / Section |
|-------|------------------------|-----------|------------------|
| 1.    | Rudra Rajpurohit       | 230911570 | IT / B           |
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### Project Title:

#### **Detection of Fake and AI-Generated Product Reviews Using Deep Learning: A Comparative Study of Four Architecture Families.**

Online product reviews are one of the key inputs to purchase decisions and the usefulness of those reviews is predicated upon the assumption that they were indeed written by people who tried the product. This is no longer necessarily a safe assumption. Review farms have been around for years but large language models make generating fluent and product specific reviews easy and virtually free. According to existing studies, human reviewers do not perform any better than random in distinguishing between reviews created using a large language model and genuine reviews, which makes automatic detection the only plausible tool available to ecommerce websites to filter out fake reviews.

The present project looks into detection of fake reviews as a text classification task and provides a fair comparison of four deep learning architectures using the same dataset, the same train-test split and the same evaluation metrics. The main dataset used in this study is Fake Reviews Dataset released by Salminen et al. (2022), which comprises 20,000 genuine Amazon reviews along with 20,000 reviews generated using a fine-tuned GPT-2 model across ten categories of products. Second test dataset used is Out-Of-Domain Deceptive Opinion Spam Corpus provided by Ott et al. (2011, 2013) and comprising 1,600 human-generated truthful and deceptive hotel reviews. This dataset allows to assess if the detector of fake product reviews performs well on human-written deceptive reviews in a different field.

Four models are trained, one per each of the four architectures. First model is classical baseline that takes TF-IDF character and word n-gram features as input for a multilayer perceptron and serves as a lower bound showing what proportion of the task can be solved using lexical statistics alone. Second model is a convolutional network over word embeddings with several filters of different widths which allows extracting locally structured patterns like templates for generic praise. Third model is a bidirectional LSTM with an attention pooling layer. This model learns sequential context and shows which tokens contribute to making decisions. Fourth model is a fine-tuned DistilBERT encoder, which is a pre-trained transformer-based encoder that adds linguistic knowledge to the task.

The four models have one common preprocessing pipeline, one common 70/15/15 train/validation/test split with fixed random seed, and one common evaluation script for measuring accuracy, precision, recall, macro F1 score and ROC-AUC. Training parameters and inference time are recorded together with accuracy in order to ensure that the comparison takes into account not only prediction performance

but deployment cost as well. The deliverables include a single comparison table for all four models, error analysis of common failures and an explanation of the differences between the architectures, specifically if there is any necessity for pre-trained contextual representations or shallower models suffice.

**Dataset(s) Identified:**

- Primary: Fake Reviews Dataset (Salminen et al., 2022), available on OSF and Kaggle. It contains approximately 40,000 labelled reviews, evenly split between 20,000 original human-written Amazon reviews (label OR) and 20,000 computer-generated reviews (label CG) produced by a GPT-2 model fine-tuned on the same corpus. Fields include product category, star rating, label and review text, spanning ten Amazon categories such as Books, Clothing, Electronics, Home and Kitchen, and Toys and Games. The classes are balanced, so accuracy and macro F1 are both meaningful.
- Secondary, used for cross-domain evaluation: Deceptive Opinion Spam Corpus v1.4 (Ott et al., 2011, 2013). It contains 1,600 hotel reviews for 20 Chicago hotels, made up of 800 truthful reviews collected from TripAdvisor and other review communities and 800 deceptive reviews written by Amazon Mechanical Turk workers, balanced across positive and negative sentiment.
- Common protocol: stratified 70/15/15 split of the primary dataset with a fixed seed, identical for every model. The secondary dataset is never used for training and serves only as a held-out transfer test.

**Proposed Models for Comparison (3 to 4, spanning different architecture families):**

| Sl. No. | Model                                                           | Architecture Family          | Rationale                                                                                                             |
|---------|-----------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1       | TF-IDF features with a multilayer perceptron                    | Classical / feed-forward     | Lower bound that measures how far surface lexical statistics alone go, with no notion of word order.                  |
| 2       | Text-CNN over trainable word embeddings (filter widths 3, 4, 5) | Convolutional                | Captures local phrase-level cues such as repeated generic praise, at low training cost.                               |
| 3       | Bidirectional LSTM with attention pooling                       | Recurrent                    | Models sequential and longer-range dependencies, and the attention weights expose the tokens driving each prediction. |
| 4       | Fine-tuned DistilBERT                                           | Transformer / self-attention | Pre-trained contextual representation, the family that defines the current state of the art in this task.             |

**Model-to-Member Assignment:**

| Member           | Reg. No.  | Model Owned                       | Shared Responsibility                                   |
|------------------|-----------|-----------------------------------|---------------------------------------------------------|
| Rudra Rajpurohit | 230911570 | Fine-tuned DistilBERT             | Evaluation harness, metric scripts and comparison table |
| Vedant Totla     | 230911526 | Bidirectional LSTM with attention | Cross-domain transfer experiment and error analysis     |

| Member                 | Reg. No.  | Model Owned     | Shared Responsibility                                              |
|------------------------|-----------|-----------------|--------------------------------------------------------------------|
| Ishan Satyanand Thakur | 230953356 | Text-CNN        | Preprocessing and tokenisation pipeline, embedding setup           |
| Vedant Agarwal         | 230953312 | TF-IDF with MLP | Dataset acquisition, exploratory analysis, literature review table |

### Existing SOTA Methods:

- Ott et al. (2011) established the standard benchmark for deceptive opinion spam and reached close to 90 percent accuracy on their hotel review corpus using a linear SVM over n-gram and psycholinguistic features, while untrained human judges performed near chance.
- Ren and Zhang (2016) showed that neural sequence models with attention over sentence representations outperform hand-engineered feature pipelines on the same benchmark, establishing recurrent architectures as a strong reference point.
- Salminen et al. (2022) generated fake reviews with GPT-2 and reported that a fine-tuned RoBERTa classifier reached about 97 percent accuracy in separating generated from original reviews, far above the accuracy of human raters on the same samples. This is the reference result the project targets with a lighter transformer.
- Zero-shot detectors of machine-generated text, such as the curvature-based DetectGPT method of Mitchell et al. (2023), form a parallel line of work that requires no labelled data but assumes access to the log probabilities of a generating model, an assumption that does not hold for review platforms.

### Objectives:

- To quantify the relative detection performance of four deep learning architecture families, namely feed-forward, convolutional, recurrent and self-attention, on fake and AI-generated product reviews under a single common evaluation protocol.
- To characterise the trade-off between detection accuracy and computational cost, measured as parameter count, training time and inference latency, for each of the four architectures.
- To determine the extent to which detectors trained on machine-generated product reviews retain accuracy on human-written deceptive reviews from a different domain.

### Importance of the Proposed Project:

Review fraud has a direct economic effect. Inflated ratings redirect consumer spending toward inferior products, penalise honest sellers, and erode confidence in the platform itself once buyers realise they were misled. Regulators in several jurisdictions have moved recently to prohibit the sale, purchase and publication of fake consumer reviews, which places a compliance obligation on platforms to detect them at scale. The technical situation has also changed. Detection research built on crowd-sourced deceptive text now has to contend with reviews written by large language models, which are fluent, specific and produced in volume, and which published studies show human moderators cannot reliably identify. Manual moderation therefore does not scale, and platforms need automated classifiers whose accuracy and cost characteristics are understood. From a course standpoint, the task is well suited to a controlled architecture comparison because the input is short text, the labels are reliable, the classes are balanced, and every architecture family covered in the syllabus has a natural formulation on it, which makes the differences in results attributable to the architecture rather than to the data.

### Preliminary Reference List:

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- [4] Y. Kim, "Convolutional neural networks for sentence classification," in Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, 2014, pp. 1746-1751.
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- [6] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems (NIPS), Long Beach, CA, USA, 2017, pp. 5998-6008.
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- [9] Y. Ren and Y. Zhang, "Deceptive opinion spam detection using neural network," in Proc. 26th Int. Conf. Computational Linguistics (COLING), Osaka, Japan, 2016, pp. 140-150.
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#### Work Plan:

| Sl. No. | Component / Work Element / Milestone                                                                 | Expected Completion (Day/Month) |
|---------|------------------------------------------------------------------------------------------------------|---------------------------------|
| 1       | Team formation, problem finalisation and synopsis submission                                         | 25 Aug 2026                     |
| 2       | Dataset acquisition, exploratory data analysis and class balance check                               | 01 Sep 2026                     |
| 3       | Shared preprocessing pipeline (cleaning, tokenisation, encoding) and fixed 70/15/15 stratified split | 08 Sep 2026                     |
| 4       | Common evaluation harness with accuracy, precision, recall, macro F1 and ROC-AUC                     | 12 Sep 2026                     |
| 5       | Literature review of 8 to 10 papers compiled into the summary table                                  | 18 Sep 2026                     |
| 6       | TF-IDF with MLP baseline and Text-CNN implemented with preliminary results                           | 22 Sep 2026                     |
| 7       | Interim report and individual contribution log submitted                                             | 25 Sep 2026                     |
| 8       | Bidirectional LSTM with attention implemented and evaluated                                          | 06 Oct 2026                     |

| <b>Sl. No.</b> | <b>Component / Work Element / Milestone</b>                                   | <b>Expected Completion (Day/Month)</b> |
|----------------|-------------------------------------------------------------------------------|----------------------------------------|
| 9              | DistilBERT fine-tuning completed and evaluated                                | 13 Oct 2026                            |
| 10             | Hyperparameter tuning, cross-domain transfer test and error analysis          | 20 Oct 2026                            |
| 11             | Comparison table, final report draft, plagiarism and AI content check, slides | 25 Oct 2026                            |
| 12             | In-class presentation and viva                                                | 26 to 29 Oct 2026                      |
| 13             | Final report and GitHub repository link submitted                             | 31 Oct 2026                            |